

LEC 1

1. FIELD THEORY

1.1. Field extension.

Definition 1.1. Let E be a field. A **subfield** F of E is a subset of E that is a field with respect to the field operations inherited from E . In this case, E is called an **extension** of F , denoted by E/F .

Alternatively, if E and F are fields and there is a monomorphism $F \rightarrow E$, then E/F is a field extension.

Example 1.2. $\mathbb{Q}[x]/(x^2 + 1) \rightarrow \mathbb{C}$, where $x \mapsto \pm\sqrt{-1}$.

Example 1.3. Let F be a field. Consider the natural ring homomorphism $f: \mathbb{Z} \rightarrow F$. Then $\ker(f) = (0)$ or $\ker(f) = (p)$, with p prime. In the first case we have $\mathbb{Q} \hookrightarrow F$ and F is called of **characteristic** 0; in the second case $\mathbb{Z}/p\mathbb{Z} \hookrightarrow F$, and F is called of **characteristic** p . Denote the characteristic of F by $\text{char}(F)$. Call $\mathbb{Q}$ and $\mathbb{Z}/p\mathbb{Z}$ the **prime subfield** of F .

Suppose have E/F . Let $S \subset E$. Denote by $F(S)$ the subfield of E generated by S : that is, the intersection of all subfields of E that contains S .

Definition 1.4. If $S = \{u\}$ is a single element, then $F(u)$ is called a **simple extension** of F , and u is called a **primitive element**.

One can write $F(u)$ as $F(u) = \left\{ \frac{f(u)}{g(u)} \mid f(x), g(x) \in F[x], g(u) \neq 0 \right\}$. Let $F[u]$ be the subring of E generated by F and u . Consider the ring homomorphism $\varphi: F[x] \rightarrow E$ such that $x \mapsto u$. Then $\text{Im } \varphi = F[u]$. If $\ker \varphi = 0$, then u is called a **transcendental element** over F ; otherwise an **algebraic element**. In the latter case suppose $\ker \varphi = (g(x))$ with monic $g(x)$, then $g(x)$ is called the **minimal polynomial** of u . It follows that $F(u) \cong F[x]/(g(x))$.

Example 1.5. $\sqrt{2}$ is an algebraic element over $\mathbb{Q}$, with minimal polynomial $x^2 - 2$.

Corollary 1.6 (Kronecker). *Suppose F is a field and $f(x) \in F[x]$. Then there is an extension E/F such that $f(x)$ has a root in E .*

Definition 1.7. Suppose E/F . If every $u \in E$ is an algebraic element, then E/F is called an **algebraic extension**; otherwise called a **transcendental extension**.

Example 1.8. $\mathbb{Q}(\sqrt{-1})/\mathbb{Q}$ is an algebraic extension, $\mathbb{Q}(\pi)/\mathbb{Q}$ is a transcendental extension.

Definition 1.9. Suppose have E/F , then E can be regarded as a vector space over F . Let $[E : F]$ denote the dimension of E over F , and called the **degree** of the extension. If $[E : F] < \infty$, call E a **finite extension** of F , otherwise **infinite extension**.

Example 1.10. Degree 2 (finite) extensions: $\mathbb{C}/\mathbb{R}$, $\mathbb{Q}(\sqrt{-1})/\mathbb{Q}$. In general, if $\text{char}(F) \neq 2$ and $[E : F] = 2$, then $E = F(a)$, where $a^2 \in F$.

Definition 1.11. Suppose E/F is a finite extension. Given $\alpha \in E$, the **norm/trace** of α is the determinant/trace of the F -linear operator $E \rightarrow E$ induced by multiplication of α , denoted by $N_{E/F}(\alpha)/\text{Tr}_{E/F}(\alpha)$.

Proposition 1.12. Suppose $a, b \in F$ and $\alpha, \beta \in E$. Then $N_{E/F}(\alpha\beta) = N_{E/F}(\alpha)N_{E/F}(\beta)$, $N_{E/F}(a\alpha) = a^{[E:F]}N_{E/F}(\alpha)$, $\text{Tr}(a\alpha + b\beta) = a\text{Tr}(\alpha) + b\text{Tr}(\beta)$. In particular, $N_{E/F}(a) = a^{[E:F]}$ and $\text{Tr}(a) = [E : F]a$.

Proposition 1.13 ([?, Section 0BIE]). Suppose $E/K/F$ are finite extensions. Then $\text{Tr}_{E/F} = \text{Tr}_{K,F} \circ \text{Tr}_{E,K}$ and $N_{E/F} = N_{K,F} \circ N_{E,K}$.

Proposition 1.14. Suppose E/F , and $u \in E$ is a algebraic element with minimal polynomial $g(x)$. Then $[F(u) : F] = \deg g(x)$. Conversely, if $[F(u) : F] < \infty$, then u is an algebraic element.

Example 1.15. Infinite extension: $\mathbb{R}/\mathbb{Q}$ (not even algebraic).

Corollary 1.16. Finite extensions are algebraic extensions.

Theorem 1.17. Suppose have $E/K/F$. Then E/F is finite if and only if both E/K and K/F are finite. In this case we have $[E : F] = [E : K] \cdot [K : F]$.

Example 1.18. $[\mathbb{Q}(\sqrt[4]{2}) : \mathbb{Q}] = 4 = 2 \cdot 2 = [\mathbb{Q}(\sqrt[4]{2}) : \mathbb{Q}(\sqrt{2})] \cdot [\mathbb{Q}(\sqrt{2}) : \mathbb{Q}]$

Proposition 1.19. The extension E/F is finite if and only if there exists algebraic elements $u_1, \dots, u_n$ such that $E = F(u_1, \dots, u_n)$.

Corollary 1.20. Suppose have $E/K/F$ and $[E : F] < \infty$. Then $[E : K] \mid [E : F]$ and $[K : F] \mid [E : F]$. In particular, if $[E : F]$ is prime, then there is no K such that $F \subsetneq K \subsetneq E$.

In general, if $F \subset K \subset E$ then K is called an **intermediate field** of E/F .

Theorem 1.21 (Steinitz). Let E/F be a finite extension. Then it is simple extension if and only if it has finitely many intermediate field.

Example 1.22. Extension of fields of prime degree must be simple.

Remark 1.23. Later: if $\text{char}(F) \neq 0$ (or more generally separable), and E/F is finite, then it is simple extension.

1.2. Algebraic extension.

Theorem 1.24. Suppose both E/F and K/E are algebraic, then K/F is algebraic.

Theorem 1.25. Suppose have E/F . Let K be the subset of E consisting of all algebraic elements over F . Then K is a field.

Example 1.26. All algebraic elements in $\mathbb{R}$ over $\mathbb{Q}$ form an algebraic extension of $\mathbb{Q}$ which is not a finite extension.

Corollary 1.27. Suppose E/F . Let $K \subset E$ be the subfield of all algebraic elements. Then any algebraic element over K is contained in K .

Definition 1.28. Suppose E/K . If there is no intermediate field K' such that K'/K is algebraic and $K \subsetneq K'$, then K is called **algebraically closed** in E . Suppose $E/K/F$, where K/F is algebraic and K is algebraically closed in E . Then K is called the **algebraic closure** of F in E .

Proposition 1.29. Suppose E/F . Then the algebraic closure of F in E consists of all algebraic element over F .

Definition 1.30. Let K be a field. If K only has trivial algebraic extension, then K is called **algebraically closed**.

Example 1.31. Consider $\mathbb{C}/\mathbb{R}/\mathbb{Q}$, where $\mathbb{C}$ is algebraically closed. The algebraic closure of $\mathbb{R}$ in $\mathbb{C}$ is just $\mathbb{C}$. The algebraic closure K of $\mathbb{Q}$ in $\mathbb{R}$ is a proper subfield of $\mathbb{R}$. By definition, K is algebraically closed in $\mathbb{R}$ but K is not an algebraically closed field.

Example 1.32. Any field F is algebraically closed in the extension $F(x)/F$.

Definition 1.33. Suppose K/F . If K is algebraic over F and K is algebraically closed, then K is called the **algebraically closure** of F .

Theorem 1.34 (To be proved later). Any field F has an algebraic closure, and the algebraic closure is unique up to an isomorphism over F .

Example 1.35. $\mathbb{C}$ is the algebraically closure of $\mathbb{R}$. The algebraically closure of $\mathbb{Q}$ in $\mathbb{R}$ is not the algebraically closure of $\mathbb{Q}$.

Theorem 1.36. Let K be a field, the following are equivalent:

- (1) K is algebraically closed
- (2) any irreducible polynomial in $K[x]$ has degree 1
- (3) any polynomial in $K[x]$ of positive degree is a product of linear polynomial
- (4) any polynomial in $K[x]$ of positive degree has at least one root in K .

Theorem 1.37. Let K be an algebraically closed field, and F a subfield of K . Then the algebraic closure $\overline{F}$ of F in K is algebraically closed, and it is the algebraic closure of F .

Given E/F . Let $\text{End}(E/F)$ and $\text{Aut}(E/F)$ be the set of endomorphisms/automorphisms from E to E that preserves elements in F . Later $\text{Aut}(E/F)$ will be denoted by $\text{Gal}(E/F)$ due to its group structure.

Proposition 1.38. Suppose E/F is algebraic. Then $\text{End}(E/F) = \text{Aut}(E/F)$.

1.3. Ruler-compass construction. Consider n points $z_1, \dots, z_n \in \mathbb{C}$. A **ruler-compass** construction from $S_0 = \{z_i\}$ is the process of marking a point in the (transverse) intersection of two of the following:

- (1) a line passing through two distinct points in S_0 ;
- (2) a circle centered at a point in S_0 and with radius given by the distance of two distinct points in S_0 .

Proposition 1.39. *Let $S \subset \mathbb{C}$ be the subset of points that can be obtained from $\{z_i\}$ by a finite sequence of ruler-compass construction. Then*

- (1) S is a subfield of $\mathbb{C}$
- (2) for any $a \in S$, we have $\sqrt{a} \in S$ and $\bar{a} \in S$, where $\bar{a}$ is the conjugation
- (3) S is the smallest subfield of $\mathbb{C}$ that contains $\{z_i\}$ and satisfies (1) and (2).

Definition 1.40. Points in S are called **constructible** from $z_1, \dots, z_n$.

Definition 1.41. Let F be a subfield of $\mathbb{C}$. An extension of F is called a **square root tower** over F if it is of the form $F(u_1, \dots, u_n)$, where $u_1^2 \in F$ and $u_i^2 \in F(u_1, \dots, u_{i-1})$ for $i > 1$.

Theorem 1.42. *Given $z_1, \dots, z_n \in \mathbb{C}$. Let $F = \mathbb{Q}(z_1, \bar{z}_1, \dots, z_n, \bar{z}_n)$. Then a point $z \in \mathbb{C}$ is constructible if and only if it is contained in some square root tower over F .*

Corollary 1.43. *Given $z_1, \dots, z_n \in \mathbb{C}$. Let $F = \mathbb{Q}(z_1, \bar{z}_1, \dots, z_n, \bar{z}_n)$. If $z \in \mathbb{C}$ is constructible, then z is algebraic over F and $[F(z) : F] = 2^k$.*

Example 1.44. Trisection of angles; duplication of the cube; Construction of regular n -gons ($n = 2^k p_1 \dots p_s$, where p_i are Fermat primes).